

## Exercise JA

Q1. A value of  $b$  for which equations  $x^2 + bx - 1 = 0$  and  $x^2 + x + b = 0$  have one root in common is

- (A)  $-\sqrt{2}$  (B)  $-i\sqrt{3}$  (C)  $i\sqrt{3}$  (D)  $\sqrt{2}$

Ans. B

Sol.  $x^2 + bx - 1 = 0$

$$\begin{array}{r} x^2 + bx - 1 = 0 \\ x^2 + x + b = 0 \\ \hline \end{array}$$

$$\frac{x^2}{b^2 + 1} = \frac{x}{-1 - b} = \frac{1}{1 - b}$$

$$\Rightarrow x = \frac{b^2 + 1}{-(b + 1)} = \frac{-(b + 1)}{1 - b} \Rightarrow (b^2 + 1)(1 - b) = (b + 1)^2$$

$$\Rightarrow b^2 - b^3 + 1 - b = b^2 + 2b + 1 \Rightarrow b^3 + 3b = 0 \Rightarrow b = 0 ; b^2 = -3 \Rightarrow b = \sqrt{3} i,$$

Q2. The quadratic equation  $p(x) = 0$  with real coefficients has purely imaginary roots. Then the equation  $p(p(x)) = 0$  has :

- (A) only purely imaginary roots (B) all real roots  
(C) two real and two purely imaginary roots (D) neither real nor purely imaginary roots

Ans. D

Sol.  $p(x)$  will be of the form  $ax^2 + c$ . Since it has purely imaginary roots only.

Since  $p(x)$  is zero at imaginary values while  $ax^2 + c$  takes real value only at real 'x', no root is real.

Also  $p(p(x)) = 0 \Rightarrow p(x)$  is purely imaginary  $\Rightarrow ax^2 + c = \text{purely imaginary}$

Hence  $x$  can not be purely imaginary since  $x^2$  will be negative in that case and  $ax^2 + c$  will be real.

Thus (D) is correct.

Q3. Let  $S$  be the set of all non-zero numbers  $\alpha$  such that the quadratic equation  $\alpha x^2 - x + \alpha = 0$  has two distinct real roots  $x_1$  and  $x_2$  satisfying the inequality  $|x_1 - x_2| < 1$ .

Which of the following intervals is (are) a subset (s) of  $S$  ?

- (A)  $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$  (B)  $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$  (C)  $\left(0, \frac{1}{\sqrt{5}}\right)$  (D) Both (A) and (B)

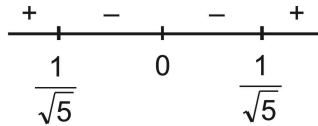
Ans. D

Sol.  $(x_1 + x_2)^2 - 4x_1x_2 < 1$

$$\frac{1}{\alpha^2} - 4 < 1$$

$$\Rightarrow 5 - \frac{1}{\alpha^2} > 0$$

$$\frac{5\alpha^2 - 1}{\alpha^2} > 0$$



$$\alpha \in \left(-\infty - \frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}\right) \dots(1)$$

$$\alpha \in \left(-\frac{1}{2}, \frac{1}{2}\right) \dots(2)$$

(1) & (2)

$$\alpha \in \left(-\frac{1}{2}, \frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$$

### Paragraph

Let  $p, q$  be integers and let  $\alpha, \beta$  be the roots of the equation,  $x^2 - x - 1 = 0$ , where  $\alpha \neq \beta$ . For  $n = 0, 1, 2, \dots$ , let  $a_n = p\alpha^n + q\beta^n$ .

**FACT** : if  $a$  and  $b$  are rational numbers and  $a + b\sqrt{5} = 0$ , then  $a = 0 = b$ .

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Q4. If  $a_4 = 28$ , then  $p + 2q =$

(A) 12 (B) 21 (C) 14 (D) 7

Ans. A

Sol.  $a_{n+2} = a_{n+1} + a_n$

$$a_4 = a_3 + a_2 = 3a_1 + 2a_0 = 3pa + 3qb + 2(p + q)$$

$$\text{As } \alpha = \frac{1 + \sqrt{5}}{2}, \beta = \frac{1 - \sqrt{5}}{2}, \text{ we get}$$

$$a_4 = 3p\left(\frac{1 + \sqrt{5}}{2}\right) + 3q\left(\frac{1 - \sqrt{5}}{2}\right) + 2p + 2q = 28$$

$$\left(\frac{3p}{2} + \frac{3q}{2} + 2p + 2q - 28\right) = 0 \dots (i)$$

$$\text{and } \frac{3p}{2} - \frac{3q}{2} = 0 \dots (ii)$$

$$p = q \text{ (from (ii))}$$

$$7p = 28 \text{ (from (i) and (ii))}$$

$$p = 4$$

$$q = 4$$

$$p + 2q = 12$$

### Paragraph

Let  $p, q$  be integers and let  $\alpha, \beta$  be the roots of the equation,  $x^2 - x - 1 = 0$ , where  $\alpha \neq \beta$ . For  $n = 0, 1, 2, \dots$ , let  $a_n = p\alpha^n + q\beta^n$ .

**FACT :** if  $a$  and  $b$  are rational numbers and  $a + b\sqrt{5} = 0$ , then  $a = 0 = b$ .

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Q5.  $a_{12} =$

- (A)  $a_{11} + 2a_{10}$  (B)  $a_{11} + a_{10}$  (C)  $a_{11} - a_{10}$  (D)  $2a_{11} + a_{10}$

Ans. B

Sol. As  $a$  and  $b$  are roots of equation  $x^2 - x - 1 = 0$ , we get :

$$a^2 - a - 1 = 0 \Rightarrow a^2 = a + 1$$

$$b^2 - b - 1 = 0 \Rightarrow b^2 = b + 1$$

$$\begin{aligned} \therefore a_{11} + a_{10} &= pa^{11} + qb^{11} + pa^{10} + qb^{10} \\ &= pa^{10}(a + 1) + qb^{10}(b + 1) \\ &= pa^{10} \times a^2 + qb^{10} \times b^2 \\ &= pa^{12} + qb^{12} \\ &= a_{12} \end{aligned}$$

Q6. Suppose  $a, b$  denote the distinct real roots of the quadratic polynomial  $x^2 + 20x - 2020$  and suppose  $c, d$  denote the distinct complex roots of the quadratic polynomial  $x^2 - 20x + 2020$ . Then the value of  $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$  is :

- (A) 0 (B) 8000 (C) 8080 (D) 16000

Ans. D

Sol.  $x^2 + 20x - 2020 = 0$   $a, b$  are its roots

$x^2 - 20x + 2020 = 0$   $c, d$  are its roots

$$a + b = -20 \quad ab = -2020$$

$$c + d = 20 \quad cd = 2020$$

$$\begin{aligned} E &= ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d) \\ &= a^2c - ac^2 + a^2d - ad^2 + b^2c + b^2d - bd^2 \\ &= a^2(c + d) + b^2(c + d) - c^2(a + b) - d^2(a + b) \\ &= (a^2 + b^2)(c + d) - (a + b)(c^2 + d^2) \\ &= ((a + b)^2 - 2ab)(c + d) - (a + b)((c + d)^2 - 2cd) \\ &= (400 + 4040)(20) - (-20)(400 - 4040) \\ &= 88800 - 72800 = 16000 \\ &= (400 + 4040)(20) - (-20)(400 - 4040) \\ &= 88800 - 72800 = 16000 \end{aligned}$$